Local and Latent: Dual Views of Symmetry Breaking in Cooperative Multi-Robot RL

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Abstract Spatial symmetry is among the most effective inductive biases for robot learning, but in real multi-robot deployments it almost never holds exactly. Violations come in two structurally distinct forms: *local* breakers, visible in the current state but confined to small regions (e.g., fixed obstacles), and *latent* breakers, hidden from observation but fixed across an episode (e.g., wind or gravity). The two are orthogonal—state-dependent gating cannot find an unobserved breaker, and hidden-context inference cannot represent localized agent-dependent violations—yet realistic deployments exhibit both. We propose a dual-view theory and unified architecture. View 1 formalizes a locally factored partially symmetric Markov game and shows that critic-level symmetry error is controlled by discounted future breaker exposure, motivating a TD-learned per-agent gated critic. View 2 formalizes a latent symmetry-breaker POMDP and bounds value error by breaker-estimation error. The unified architecture pairs an equivariant latent encoder with per-agent gating.

Keywords cooperative multi-robot RL, group equivariance, symmetry breaking

1 Introduction

Group symmetries—geometric transformations leaving system properties invariant—provide one of the most effective inductive biases in robot learning, with consistent gains in imitation learning [1, 2, 3] and reinforcement learning (RL) [4, 5, 6], where exploration is expensive and demonstrations are scarce. Formally, given a group G acting on the state space $\mathcal{S}$ and action space $\mathcal{A}$ of an MDP, the MDP is G-symmetric if for every $g \in G$,

$$P(g \cdot s' | g \cdot s, g \cdot a) = P(s' | s, a), \quad R(g \cdot s, g \cdot a) = R(s, a). \tag{1}$$

The notion extends to multi-robot settings by lifting actions and invariance to joint spaces (PI's [7]), where the relevant group factors as $G = G_{\text{spatial}} \times \mathcal{S}_n$. We focus on the spatial component and leave permutation symmetry for future work.

In practice, condition (1) almost never holds exactly. We observe that violations fall into two structurally distinct modes, separated by a single diagnostic: *is the breaker present in an agent's current observation, or hidden from it?* *Local* breakers are observable and confined to small regions of the per-agent state-action space—e.g., for warehouse robots near a fixed shelving unit, symmetry breaks only in the unit's vicinity, and at any moment only a subset of agents is affected. *Latent* breakers are hidden global variables fixed within an episode—e.g., wind on aerial drones, gravity on inclined multi-arm setups, current on underwater robots—where dynamics would be exactly symmetric *if* the vector were rotated together with state and action, yet the vector is not directly observable.

The two modes are structurally orthogonal: local methods cannot detect hidden breakers; latent inference cannot represent agent-dependent localized violations. Realistic deployments exhibit both. This *local–latent* dichotomy separates our proposal from prior work, which either treats single-agent settings where the dichotomy does not arise or globally relaxes symmetry without distinguishing structurally different modes [8, 9, 10, 11, 12, 13]. Closing this gap is the goal of this proposal.

2 Methods

We consider a discounted cooperative Markov game $\mathcal{M} = \langle \mathcal{N}, \mathcal{S}, \mathcal{A}, P, r, \gamma \rangle$ with agent set $\mathcal{N} = \{1, \dots, N\}$, joint action space $\mathcal{A} = \mathcal{A}^1 \times \cdots \times \mathcal{A}^N$, and a spatial group G lifted to each agent's state and action via group actions $(L_g^i, K_g^i)\colon \mathcal{X}^i \times \mathcal{A}^i \to \mathcal{X}^i \times \mathcal{A}^i$. We pursue two complementary views of why G may fail to be exact: *View 1* attributes failures to localized regions of the observed local state-action space, motivating a state-dependent gated critic; *View 2* attributes them to an unobserved global breaker variable (wind, gravity, current) and recovers equivariance in the augmented state through breaker inference. The two are combined in a single architecture.

2.1 View 1: symmetry breakers as locally observed state-action regions

We posit a *locally factored partially symmetric Markov game* in which the joint state, reward, and transition decompose agent-wise: $\mathcal{S} = \mathcal{X}^1 \times \cdots \times \mathcal{X}^N, r(s,a) = \sum_{i=1}^N r_i(x^i, a^i), P(\cdot \mid s, a) = \bigotimes_{i=1}^N P_i(\cdot \mid x^i, a^i)$. Each local factor carries its own finite symmetry group G_i acting on $(\mathcal{X}^i, \mathcal{A}^i)$ and its own *local breaker region* $\mathcal{B}^i \subseteq \mathcal{X}^i \times \mathcal{A}^i$. On the safe set $(x^i, a^i) \notin \mathcal{B}^i$ the local reward and transition are G_i-invariant; on $\mathcal{B}^i$ they may be arbitrary. The breaker region is itself G_i-invariant, so the safe/broken designation is a property of the orbit. This is more general than a single global gate over the joint state: symmetry can be exploited at agents whose local context is symmetric even when others' are not.

Theory. We will establish two results. First, define a *routed model* that, on each factor, uses the group-averaged reward and transition on the safe set and the raw versions on the breaker region:

$$\begin{aligned} \widetilde{r}_i(x^i, a^i) &:= (1 - \lambda_{\text{cur}}^i)\, \overline{r}_i(x^i, a^i) + \lambda_{\text{cur}}^i\, r_i(x^i, a^i), \\ \widetilde{P}_i &:= (1 - \lambda_{\text{cur}}^i)\, \overline{P}_i + \lambda_{\text{cur}}^i\, P_i, \end{aligned} \tag{3}$$

with $\lambda_{\text{cur}}^i(x^i, a^i) := \mathbf{1}\{(x^i, a^i) \in \mathcal{B}^i\}$ and $\overline{r}_i, \overline{P}_i$ the G_i-orbit averages. We will show the routed Bellman operator $\widetilde{\mathcal{T}}$ coincides with the true $\mathcal{T}$, so $\widetilde{Q}^* = Q^*$: oracle agent-wise routing is exact at the one-step model level.

One-step model symmetry does *not* imply critic symmetry, because the critic accumulates discounted future rewards along trajectories that may enter breaker regions. We will prove that for a safe-equivariant local policy π_i and any safe pair $(x, a) \notin \mathcal{B}^i$,

$$\left| Q_i^{\pi_i}(L_g^i[x], K_g^i[a]) - Q_i^{\pi_i}(x, a) \right| \;\le\; 2\, V_{i,\max}\, \beta_i^{\pi_i}(x, a), \tag{4}$$

where $\beta_i^{\pi_i}(x, a) := \mathbb{E}_{x,a}^{\pi_i}[\gamma^{\tau_i}]$ is the *discounted future breaker exposure* and τ_i is the first hitting time of $\mathcal{B}^i$. A finite-horizon variant replaces β with $\eta_{i,H}^{\pi_i}(x, a) + \gamma^{H+1}$. Symmetry bias on the critic is low exactly where future breaker exposure is small.

Algorithm: TD-learned gated critic. The future-exposure bound motivates gating at the *critic level* by mixing two outputs: an orbit-averaged invariant critic $Q_\theta^{\text{inv}}(s, a) = \frac{1}{|G|} \sum_{g \in G} f_\theta(g \cdot s, g \cdot a)$, an unconstrained raw critic $Q_\phi^{\text{raw}}(s, a)$, and a learned gate $\widehat{\lambda}_\psi(s, a) = \sigma(g_\psi(s, a)) \in [0, 1]$, combined as $Q_{\theta,\phi,\psi}^{\text{gate}}(s, a) = (1 - \widehat{\lambda}_\psi)\, Q_\theta^{\text{inv}} + \widehat{\lambda}_\psi\, Q_\phi^{\text{raw}}$. A local-factor variant sums per-agent gated branches, matching the future-exposure bound more directly. The gate is trained *purely* via TD loss—no auxiliary supervision on breaker labels. The gradient $\partial Q^{\text{gate}} / \partial \widehat{\lambda}_\psi = Q_\phi^{\text{raw}} - Q_\theta^{\text{inv}}$ provides the routing signal: when the raw branch reduces TD error more than the invariant branch, the gate opens; when they agree, it favors the data-efficient invariant branch. We implement this within a CTDE actor-critic with decentralized actors $a^i = \mu_i(o^i)$ and a centralized gated critic.

2.2 View 2: symmetry breakers as partially observed latent state

For hidden-context breakers, the agent-wise local-region view does not fit: the breaker is a *single global object* (e.g., a wind or gravity vector) shared by all agents and fixed across the episode. Throughout this subsection we therefore use single-agent-style notation (s, a, o, b), with the understanding that joint quantities $s = (x^1, \ldots, x^N)$, $a = (a^1, \ldots, a^N)$, $o = (o^1, \ldots, o^N)$ decompose per-agent and G acts simultaneously on all agents' frames; agent-wise factorization returns in § 2.3. We model the environment as a *latent symmetry-breaker POMDP* $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{B}, P_b, r_b, \Omega_b, \gamma, \mu_0, \rho_0 \rangle$ in which $b \sim \rho_0$ is sampled at the start of each episode and held fixed; the agent receives observations $o_t \sim \Omega_b(\cdot \mid s_t)$ and history h_t but never sees b. The structural assumption is *augmented equivariance*: for every $g \in G$,

$$r_{g\cdot b}(g\cdot s, g\cdot a) = r_b(s, a), \quad P_{g\cdot b}(g\cdot C \mid g\cdot s, g\cdot a) = P_b(C \mid s, a), \quad \Omega_{g\cdot b}(g\cdot D \mid g\cdot s) = \Omega_b(D \mid s). \tag{5}$$

The system is symmetric if and only if b is transformed alongside (s, a); the visible state alone admits no group action consistent with the dynamics.

Theory. We will establish three results. First, if the dynamics are non-trivial in b, any critic enforcing observation-only equivariance $Q(g\cdot o, g\cdot a) = Q(o, a)$ is inconsistent with the true Bellman targets at some (o, a, g); the correct invariance is $Q(g\cdot o, g\cdot a, g\cdot z) = Q(o, a, z)$ where z is a breaker-sufficient statistic of history. Second, under Lipschitz assumptions $|r_b - r_{b'}| \leq L_r d_{\mathcal{B}}(b, b')$ and $\|P_b - P_{b'}\|_{\mathrm{TV}} \leq L_p d_{\mathcal{B}}(b, b')$, conditioning on an estimate $\widehat{b}$ rather than the true b yields a fixed-point value error bounded by $\frac{L_r + \gamma V_{\max} L_p}{1-\gamma}\, d_{\mathcal{B}}(b, \widehat{b})$, motivating auxiliary objectives that drive $d_{\mathcal{B}}(b, \widehat{b})$ down. Third, under (5) and an equivariant prior ρ_0, the Bayesian posterior over b transforms equivariantly: $\beta_t^{g\cdot h_t}(g \cdot B) = \beta_t^{h_t}(B)$; the encoder f_η should therefore be built so its output transforms in the same representation as b.

Algorithm: breaker-conditioned actor-critic. The architecture has three components. A *latent breaker encoder* $z_t = f_\eta(h_t)$ (recurrent or contrastive) is built so $f_\eta(g \cdot h_t) = g \cdot f_\eta(h_t)$, i.e., its output lives in the same representation as b. A *breaker-conditioned policy and critic* are equivariant in the augmented pair (o, z): $\pi_\theta(g\cdot a \mid g\cdot o, g\cdot z) = \pi_\theta(a \mid o, z)$ and $Q_\omega(g\cdot o, g\cdot a, g\cdot z) = Q_\omega(o, a, z)$. The encoder is trained with auxiliary objectives in addition to TD: a dynamics loss $\mathcal{L}_{\mathrm{dyn}} = \mathbb{E}[\ell(D_\psi(o_t, a_t, z_t), o_{t+1})]$, a contrastive loss pulling same-episode representations together (and pushing across-episode pairs apart), and an optional supervised loss $\mathcal{L}_{\mathrm{sup}} = \mathbb{E}[d_{\mathcal{B}}(f_\eta(h_t), b)^2]$ when simulator labels are available. We will explore three variants: a *point-estimate* encoder producing $\widehat{b}_t \in \mathcal{B}$; a *belief* representation outputting distributional parameters; and *canonicalization*, rotating inputs into a frame aligned with $\widehat{b}_t$, applying a standard network, and rotating outputs back.

2.3 Unifying perspective and combined architecture

The two views compose because they operate at *different granularities*: View 1 is per-agent and state-dependent; View 2 is global and per-episode. Concretely, we replace View 1's orbit-averaged invariant branch with View 2's breaker-conditioned variant—conditioning on a shared encoder $z_t = f_\eta(h_t)$ broadcast to all agents—while keeping the per-agent gate. Each agent's critic becomes

$$Q_i(x^i, a^i, h_t) = \big(1 - \widehat{\lambda}^i_\psi(x^i, a^i)\big)\, q^{\mathrm{inv}}_{i,\theta}(x^i, a^i, z_t) + \widehat{\lambda}^i_\psi(x^i, a^i)\, q^{\mathrm{raw}}_{i,\phi}(x^i, a^i, h_t), \tag{6}$$

summed to $Q(s, a, h_t) = \sum_i Q_i$. Confirming that this combined architecture dominates both single-view variants on tasks mixing the two failure modes is a primary empirical claim.

Experimental design. We evaluate across three task families, each isolating a different combination of failure modes. *Multi-robot navigation with fixed obstacles* (cooperative goal-reaching on a

2D arena with a few fixed asymmetric obstacles) isolates agent-wise local breakers: at any timestep a subset of agents is near an obstacle while others are not; the group is $G = \mathrm{SO}(2)$ or a discrete subgroup, with symmetry-breaking strength varied by obstacle density. *Hidden wind navigation* replaces the obstacles with a per-episode constant wind vector omitted from observations, isolating hidden global breakers; wind direction must be inferred from the discrepancy between commanded and realized displacement. Finally, *combined regime* is multi-arm manipulation on an inclined surface (hidden gravity) with localized obstacles, testing the combined architecture (§ 2.3) and the orthogonality claim. Metrics follow standard practice—final return, sample efficiency, training stability across seeds—plus view-specific diagnostics: gate activation maps and their correlation with oracle breaker labels (View 1), and angular error between $\widehat{b}_t$ and the true wind/gravity vector (View 2). Performance is reported as a function of symmetry-breaking strength to characterize the regime where each view, and the combined architecture, dominates.

Baselines and ablations. On View 1 we compare against a raw MLP critic, a fully invariant critic, and a *current-breaker oracle gate* using ground-truth $\mathbf{1}\{(s,a) \in \mathcal{B}\}$—a diagnostic for whether current breaker information alone suffices (the future-exposure bound predicts it does not when trajectories cross breaker regions several steps ahead). On View 2: an unstructured MLP, a strict observation-only equivariant model, a recurrent non-equivariant model with history but no symmetry prior, an *oracle* receiving the true b (upper bound), and an *observable-breaker ablation* exposing b in the observation (confirming the latent variable is the missing information). We ultimately will compare the combined architecture against the strongest single-view variant from each.

3 Expected Results

The project spans 14 months. The first three months focus on setup and baseline implementation. View 1 theory and algorithms are developed in months 2–6, running in parallel with View 2 theory and algorithms in months 4–8. Unification of the two views begins in month 8 and continues through month 11. The final three months are dedicated to writing code, paper, and submission.

Funds needed. *Cash funding:* \$42,450 is requested for the annual stipend of a research assistant (RA) [14]. *AWS promotional credits:* \$50,000 is requested. ∼\$35,000 will be allocated to EC2 for parallelized multi-robot RL training, ∼\$10,000 to RoboMaker and SageMaker to enable realistic and large-scale robotics simulation, training, and benchmarking, ∼\$5,000 to supporting infrastructure.

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